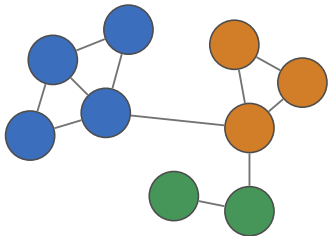


(a) Homophilous

citation / co-authorship / co-purchase
e.g. Cora, $h_{\text{edge}} = 0.81$



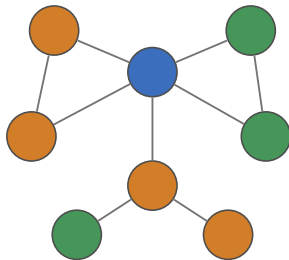
most edges join *same*-class nodes
 $\Rightarrow$ neighbourhood averaging helps



All nodes are of one type and all edges of one type, so the graph is *homogeneous* in both panels: homophily is a property of the *labels*, not of the schema.

(b) Heterophilous

web-page hyperlink / actor co-occurrence
e.g. Texas, $h_{\text{edge}} = 0.11$



edges mostly join *different*-class nodes
 $\Rightarrow$ plain GCN averaging destroys signal